## SoccerGMR Ablation Study: Stage 2 G1 Selection vs. mAP Selection (9 Backbones)

Comparing GMR Baseline vs. GMR + DQ-CGPv3 vs. GEQ with Val G1 Selection vs. GEQ with Val mAP Selection on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries).

| | | Standard Test Split (1,036 queries) | | | | Full-Decodable Test Split (2,970 queries) | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Backbone** | **Configuration** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** |
| **Moment-DETR** | 1. GMR Baseline | 6.68 | 24.28 | 67.94 | 70.20 | 5.47 | 23.48 | 72.08 | 78.91 |
| | 2. GMR + DQ-CGPv3 | 10.66 | 22.49 | 72.20 | 72.09 | 9.99 | 21.83 | 74.92 | 80.24 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | **12.64** | **15.62** | **75.48** | 73.48 | **11.63** | **15.40** | **78.93** | 80.37 |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 12.63 | 23.44 | 68.77 | **75.17** | 11.56 | 22.53 | 69.83 | **81.94** |
| **EaTR** | 1. GMR Baseline | 16.49 | 21.51 | 65.60 | 76.56 | 15.19 | 20.39 | 67.38 | 81.59 |
| | 2. GMR + DQ-CGPv3 | 16.02 | 23.87 | 65.27 | **77.19** | 15.35 | 22.31 | 65.29 | 80.38 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 17.48 | **16.31** | **75.43** | 76.17 | 15.67 | **15.48** | **78.73** | **83.04** |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | **17.49** | 22.74 | 70.12 | 76.28 | **15.98** | 22.10 | 72.31 | 82.28 |
| **QD-DETR** | 1. GMR Baseline | 14.62 | 20.23 | 71.48 | **77.01** | 12.65 | 19.05 | 72.49 | 81.75 |
| | 2. GMR + DQ-CGPv3 | 15.82 | 20.47 | 69.02 | 74.24 | 13.93 | 19.23 | 71.63 | 81.64 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | **17.28** | **18.19** | **72.25** | 75.18 | **14.20** | **17.07** | **75.61** | **82.48** |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 16.80 | 19.15 | 68.29 | 74.73 | 13.61 | 17.68 | 71.15 | 82.20 |
| **CG-DETR** | 1. GMR Baseline | **18.65** | 25.66 | 57.47 | **76.40** | **14.67** | 23.28 | 63.23 | 81.81 |
| | 2. GMR + DQ-CGPv3 | 13.49 | 22.05 | 75.04 | 74.63 | 12.49 | 20.73 | 76.97 | 79.33 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 14.75 | **15.16** | **75.31** | 75.77 | 13.17 | **14.53** | **78.87** | **82.88** |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 14.75 | 25.36 | 58.42 | 75.41 | 13.62 | 23.73 | 62.98 | 82.44 |
| **BM-DETR** | 1. GMR Baseline | 4.00 | 17.59 | **75.00** | 71.55 | 3.65 | 17.62 | 78.58 | 79.11 |
| | 2. GMR + DQ-CGPv3 | 7.74 | 17.40 | 74.43 | 70.69 | 7.32 | 16.94 | 77.84 | 79.64 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 7.82 | **12.32** | **75.00** | 72.09 | 7.32 | **12.38** | **78.80** | 79.15 |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | **8.69** | 21.58 | 71.33 | **73.71** | **7.82** | 20.87 | 75.15 | **80.15** |
| **BAM-DETR** | 1. GMR Baseline | **23.25** | 19.82 | 70.43 | **78.66** | 20.62 | 18.37 | 71.79 | 80.99 |
| | 2. GMR + DQ-CGPv3 | 21.52 | 19.40 | 71.83 | 77.80 | 19.57 | 18.04 | 72.01 | 80.13 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 22.84 | **17.60** | **74.07** | 77.41 | 20.74 | **16.77** | **77.58** | **83.32** |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 22.83 | 21.91 | 65.46 | 77.08 | **21.21** | 20.99 | 68.07 | 82.87 |
| **Sim-DETR** | 1. GMR Baseline | 20.73 | 22.51 | 74.69 | 72.97 | 19.18 | 21.78 | 75.72 | 78.87 |
| | 2. GMR + DQ-CGPv3 | 20.63 | 21.16 | 75.43 | 75.84 | 19.62 | 20.57 | 75.69 | 80.20 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 21.43 | **15.44** | **75.56** | **76.79** | 19.50 | **14.77** | **76.12** | **82.41** |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | **21.71** | 24.40 | 67.73 | 76.70 | **20.19** | 22.89 | 70.09 | 80.95 |
| **UVCOM** | 1. GMR Baseline | 7.12 | **17.24** | **75.00** | 70.03 | 6.40 | **16.83** | **76.80** | **78.98** |
| | 2. GMR + DQ-CGPv3 | 17.06 | 21.36 | 70.06 | **76.83** | 16.41 | 20.78 | 67.20 | 75.32 |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | **18.40** | 18.48 | 72.28 | 76.65 | **17.12** | 17.66 | 71.45 | 76.29 |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 17.16 | 23.49 | 59.80 | 75.83 | 16.50 | 22.37 | 59.55 | 75.48 |
| **FlashVTG** | 1. GMR Baseline | 25.43 | 25.47 | 66.35 | 74.30 | 20.43 | 24.13 | 68.48 | 82.21 |
| | 2. GMR + DQ-CGPv3 | **26.13** | 22.36 | 67.25 | **76.02** | **21.11** | 20.90 | 68.95 | **82.76** |
| | **3. GMR + DQ-CGPv3 + GEQ (G1 Sel)** | 25.81 | **17.57** | **74.62** | 74.69 | 20.87 | **16.52** | **78.13** | 81.95 |
| | **4. GMR + DQ-CGPv3 + GEQ (mAP Sel)** | 26.11 | 20.69 | 73.23 | 74.34 | 21.07 | 19.75 | 76.92 | 81.91 |

Method 3 (G1 Sel): Global Event Query fine-tuned with model selection on Val Mean G-mIoU@1 (0.4/0.5/0.6) (Official).
Method 4 (mAP Sel): Global Event Query fine-tuned with model selection on highest Val Localization mAP.
Orc-Gap (%): Area between Oracle envelope and G-mIoU@1 threshold curve across tau in [0, 1] (Integral of Oracle_G1 minus G1-AUC).
Note: Lower Orc-Gap is better (bold marks lowest gap per backbone), directly quantifying G-mIoU loss caused by imperfect query existence gating.